

1. A person driving car at a constant speed of 15 m s^{-1} is approaching a vertical wall. The person notices a change of 40 Hz in the frequency of his car's horn upon reflection from the wall. The frequency of horn is _____ Hz.
(Given: Speed of sound: 30 m s^{-1})
[2023 (06 Apr Shift 1)]
2. The ratio of speed of sound in hydrogen gas to the speed of sound in oxygen gas at the same temperature is:
[2023 (06 Apr Shift 2)]
(1) 1 : 2
(2) 4 : 1
(3) 1 : 4
(4) 1 : 1
3. The engine of a train moving with speed 10 m s^{-1} towards a platform sounds a whistle at frequency 400 Hz. The frequency heard by a passenger inside the train is: (Neglect air speed. Speed of sound in air = 330 m s^{-1})
[2023 (08 Apr Shift 1)]
(1) 400 Hz
(2) 200 Hz
(3) 412 Hz
(4) 388 Hz
4. An organ pipe 40 cm long is open at both ends. The speed of sound in air is 360 m s^{-1} . The frequency of the second harmonic is _____ Hz.
[2023 (08 Apr Shift 1)]
5. A guitar string of length 90 cm vibrates with a fundamental frequency of 120 Hz. The length of the string producing a fundamental of 180 Hz will be _____ cm
[2023 (08 Apr Shift 2)]
6. A transverse harmonic wave on a string is given by $y(x, t) = 5 \sin(6t + 0.003x)$ where x and y are in cm and t in sec. The wave velocity is _____ m s^{-1} .
[2023 (10 Apr Shift 1)]
7. The equation of wave is given by $Y = 10^{-2} \sin 2\pi \left(160t - 0.5x + \frac{\pi}{4} \right)$, where x and Y are in m and t in s. The speed of the wave is _____ km h^{-1} .
[2023 (11 Apr Shift 1)]
8. A car P travelling at 20 m s^{-1} sounds its horn at a frequency of 400 Hz. Another car Q is travelling behind the first car in the same direction with a velocity 40 m s^{-1} . The frequency heard by the passenger of the car Q is approximately [Take, velocity of sound = 360 m s^{-1}]
[2023 (11 Apr Shift 2)]
(1) 421 Hz
(2) 471 Hz
(3) 485 Hz
(4) 514 Hz
9. A wire of density $8 \times 10^3 \text{ kg m}^{-3}$ is stretched between two clamps 0.5 m apart. The extension developed in the wire is $3.2 \times 10^{-4} \text{ m}$. If $Y = 8 \times 10^{10} \text{ N m}^{-2}$, the fundamental frequency of vibration in the wire will be _____ Hz
[2023 (11 Apr Shift 2)]
10. For a certain organ pipe, the first three resonance frequencies are in the ratio of 1 : 3 : 5 respectively. If the frequency of fifth harmonic is 405 Hz and the speed of sound in air is 324 m s^{-1} the length of the organ pipe is _____ m.
[2023 (12 Apr Shift 1)]
11. In an experiment with sonometer when a mass of 180 g is attached to the string, it vibrates with fundamental frequency of 30 Hz. When a mass m is attached, the string vibrates with fundamental frequency of 50 Hz. The value of m is _____ g.
[2023 (13 Apr Shift 2)]
12. The fundamental frequency of vibration of a string between two rigid support is 50 Hz. The mass of the string is 18 g and its linear mass density is 20 g m^{-1} . The speed of the transverse waves so produced in the string is _____ m s^{-1} .
[2023 (15 Apr Shift 1)]

ANSWER KEYS

1. (420) 2. (2) 3. (1) 4. (900) 5. (60) 6. (20) 7. (1152) 8. (2)
9. (80) 10. (1) 11. (500) 12. (90)

1. (420)

The formula to calculate the frequency (f') as heard by an observer with respect to the frequency (f) produced from a source is given by

$$f' = \frac{v \pm v_o}{v \mp v_s} f \quad \dots (1)$$

where, v is the speed of sound in air, v_o is the speed of the observer and v_s is the speed of the source.

In the first situation, the source is the moving car and the observer is the wall. Hence, the frequency (f_w) as heard on the wall can be written as

$$\begin{aligned} f_w &= \frac{v-0}{v-v_c} f \\ &= \frac{v}{v-v_c} f \quad \dots (2) \end{aligned}$$

where, v_c is the speed of the car.

In the second situation, when the sound reflects back from the wall, the source of sound is the wall and the observer is the car.

Hence, the frequency (f_c) of sound as heard by the car after the reflection is given by

$$\begin{aligned} f_c &= \frac{v+v_c}{v-0} f_w \\ &= \frac{v+v_c}{v} f_w \quad \dots (3) \end{aligned}$$

From equation (2) and (3), it follows that

$$\begin{aligned} f_c &= \frac{v+v_c}{v} \times \frac{v}{v-v_c} f \\ &= \frac{v+v_c}{v-v_c} f \quad \dots (4) \end{aligned}$$

It is given in the problem that

$$f_c - f = 40 \quad \dots (5)$$

Substitute the expression from equation (4) into equation (5) and solve to calculate the frequency of the horn.

$$\begin{aligned} \frac{v+v_c}{v-v_c} f - f &= 40 \\ \Rightarrow \frac{2v_c}{v-v_c} f &= 40 \\ \Rightarrow f &= 40 \times \frac{v-v_c}{2v_c} = 40 \times \frac{330-15}{2 \times 15} \text{ m s}^{-1} = 420 \text{ m s}^{-1} \end{aligned}$$

2. (2)

The formula to calculate the speed of sound in hydrogen gas is given by

$$v_{H_2} = \sqrt{\frac{\gamma RT}{M_{H_2}}} \quad \dots (1)$$

The formula to calculate the speed of sound in oxygen is given by

$$v_{O_2} = \sqrt{\frac{\gamma RT}{M_{O_2}}} \quad \dots (2)$$

Divide equation (1) by (2) and simplify to obtain the required ratio.

$$\frac{v_{H_2}}{v_{O_2}} = \sqrt{\frac{M_{O_2}}{M_{H_2}}} \quad \dots (3)$$

Substitute the values of the known parameters into equation (3) to calculate the required ratio.

$$\begin{aligned} \frac{v_{H_2}}{v_{O_2}} &= \sqrt{\frac{32}{2}} \\ &= 4 : 1 \end{aligned}$$

3. (1)

The formula to calculate the frequency of the sound as heard by the passenger inside the train can be written as

$$f' = f_0 \left[\frac{v-v_o}{v-v_s} \right] \quad \dots (1)$$

Since the passenger is inside the train, the speeds of the source and the observer are the same.

Substitute the values of the known parameters into equation (1) to get the required frequency.

$$\begin{aligned} f' &= 400 \left[\frac{330-10}{330-10} \right] \text{ Hz} \\ &= 400 \text{ Hz} \end{aligned}$$

4. (900)

The formula for the frequency in an open organ pipe is

$$f = \frac{nv}{2L}$$

Since, it is a second harmonic, $n = 2$

The frequency of the second harmonic becomes $f = \frac{2v}{2L} = \frac{v}{L}$

The data given is

$$L = 0.4 \text{ m}$$

$$v = 360 \text{ m s}^{-1}$$

Substituting the values in the frequency formula

$$f = \frac{360 \text{ m s}^{-1}}{0.4 \text{ m}} = 900 \text{ Hz}$$

5. (60)

The formula to calculate the fundamental frequency produced in a vibrating string is given by

$$f_0 = \frac{v}{2L} \dots (1)$$

For the first case, it can be written that

$$120 = \frac{v}{2(90)} \dots (2)$$

Similarly, for the second case, it can be written that

$$180 = \frac{v}{2(L)} \dots (3)$$

Divide equation (3) by equation (2) and solve to obtain the required length of the string.

$$\frac{180}{120} = \frac{\frac{v}{2(L)}}{\frac{v}{2(90)}}$$

$$= \frac{90}{L}$$

$$\Rightarrow L = 90 \times \frac{120}{180}$$

$$= 60$$

6. (20)

The general equation for a wave can be written as $y = A \sin(\omega t + kx)$. Comparing it with the given equation, $y = 5 \sin(6t + 0.003x)$,

$$\omega = 6 \text{ rad s}^{-1}, k = 0.003 \text{ cm}^{-1}$$

Using the equation, $v = \frac{\omega}{k}$

$$v = \frac{6}{0.003 \times 10^2} = 20 \text{ m s}^{-1}$$

7. (1152)

The equation of the wave is

$$Y = 10^{-2} \sin 2\pi \left(160t - 0.5x + \frac{\pi}{4} \right)$$

The velocity of a wave is given by $v = \frac{\omega}{k} \dots (i)$.

A wave equation is given by

$$Y = A \sin(\omega t - kx + \phi)$$

Using equation (i),

$$v = \left(\frac{2\pi \times 160}{2\pi \times 0.5} \right) = 320 \text{ m s}^{-1}$$

$$\Rightarrow v = 320 \times \frac{18}{5}$$

$$\Rightarrow v = 1152 \text{ km h}^{-1}$$

8. (2)



Let velocity of sound in air be $v \text{ m s}^{-1}$. Using Doppler formula,

$$f = f_0 \left(\frac{v + v_o}{v - v_s} \right)$$

$$= f_0 \left(\frac{v + 40}{v - 20} \right)$$

$$= 400 \left(\frac{360 + 40}{340} \right)$$

$$\approx 471 \text{ Hz}$$

9. (80)

Using the relation of Young's modulus,

$$\frac{T}{A} = Y \left(\frac{\Delta L}{L} \right) \\ \Rightarrow T = \left(\frac{Y \Delta L}{L} \times A \right)$$

The linear mass density is $\mu = \left(\frac{m}{L} \right)$.

So,

$$\frac{T}{\mu} = \frac{Y \Delta L A}{L \left(\frac{m}{L} \right)} = \frac{Y(\Delta L) \times L A}{L(m)} = \left(\frac{Y \Delta L}{L} \right) \times \left(\frac{1}{\rho} \right)$$

Substituting the values,

$$\frac{T}{\mu} = \frac{8 \times 10^{10} \times 3.2 \times 10^{-4}}{0.5} \times \left(\frac{1}{8 \times 10^3} \right) = 6.4 \times 10^3$$

$$\Rightarrow \frac{T}{\mu} = 64 \times 10^2$$

The fundamental frequency is given by $f = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$.

$$\Rightarrow \sqrt{\frac{T}{\mu}} = 8 \times 10 = 80 \text{ m s}^{-1}$$

Therefore,

$$f = \left(\frac{80}{1} \right) = 80 \text{ Hz}$$

10. (1)

It is given,

$$f_5 = 405 \text{ Hz.}$$

As the first three resonance frequencies are in the ratio of 1 : 3 : 5, therefore it must be a closed organ pipe.

The frequency formula in a closed organ pipe is given by $f = \frac{nv}{4l}$. So,

$$\frac{5v}{4l} = 405 \text{ Hz}$$

$$\Rightarrow l = \frac{405 \times 4}{5 \times 324} = 1 \text{ m}$$

11. (500)

The formula to calculate the fundamental frequency in a stretched string is given by

$$f = \frac{1}{2L} \sqrt{\frac{T}{\mu}} \dots (1)$$

For the first mass, it can be written that

$$f_1 = \frac{1}{2L} \sqrt{\frac{m_1 g}{\mu}} \dots (2)$$

And, for the second mass, it can be written that

$$f_2 = \frac{1}{2L} \sqrt{\frac{m_2 g}{\mu}} \dots (3)$$

Divide equation (3) by equation (2) and simplify to obtain the required mass.

$$\frac{f_2}{f_1} = \frac{\frac{1}{2L} \sqrt{\frac{m_2 g}{\mu}}}{\frac{1}{2L} \sqrt{\frac{m_1 g}{\mu}}}$$

$$= \sqrt{\frac{m_2}{m_1}}$$

$$\Rightarrow \frac{m_1}{m_2} = \left(\frac{f_1}{f_2} \right)^2$$

$$\Rightarrow m_2 = m_1 \left(\frac{f_2}{f_1} \right)^2 \dots (4)$$

Substitute the values of the known parameters into equation (4) to calculate the required mass.

$$m_2 = 180 \text{ g} \times \left(\frac{50 \text{ Hz}}{30 \text{ Hz}} \right)^2$$

$$= 500 \text{ g}$$

12. (90)

The fundamental frequency is $f = 50 \text{ Hz}$.

The length of the string is $l = \frac{\lambda}{2}$

Linear mass density is $\mu = \frac{m}{l} = 20 \text{ g m}^{-1}$

Using $v = f\lambda$

$$\frac{v}{2l} = 50$$

$$\Rightarrow v = 100 \times l = \frac{100 \times 18}{20} = 90 \text{ m s}^{-1}$$